

Sunway Village Broadband Infrastructure Design

WISP Network Architecture & Deployment Plan

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v1.0	06 March 2026	Initial technical architecture submission

Objective: Deliver scalable, affordable broadband access to 1,500 households and up to 10 SMEs, using a data-first monitoring strategy and zero-touch provisioning.

Executive Summary

Project Overview

This document outlines the design for a fixed-wireless broadband deployment for Sunway Village, Hartbeespoort. The design targets approximately 1,500 households and up to 10 SMEs. The architecture prioritizes cost-efficient deployment while maintaining carrier-grade reliability through a hybrid MikroTik / Ubiquiti stack and a licensed 11 GHz microwave primary backhaul.

Financial Overview

- Total Project CAPEX: R620,000
- Phase 1 Target: 300 subscribers (20% penetration)
- Projected MRR at Target: R120,000
- OPEX Break-even: 12% penetration
- Projected CAPEX ROI: 8–12 months (based on 20% subscriber penetration)

Operational Differentiators

- Zero-Touch GitOps-based provisioning pipeline
- Deterministic CGNAT port allocation for compliance

- Cloud-hosted monitoring stack (InfluxDB/Grafana)
- Anomaly detection models trained on rolling 30-day traffic baselines
- Trigger-based capacity expansion when sector utilisation reaches ~70%

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Design Principles

The network architecture for the Sunway Village Smart WISP deployment is guided by several key engineering principles to ensure reliability, scalability, and operational efficiency.

- **High Availability:** Redundant backhaul links ensure connectivity remains available even during equipment failure or link degradation.
- **Network Segmentation:** VLAN-based segmentation isolates management, backhaul, and client traffic to enhance security and simplify troubleshooting.
- **Scalable IP Architecture:** The addressing plan is designed to allow future expansion of access points, sectors, and subscriber growth without major reconfiguration.
- **Automation-Driven Operations:** Configuration templating and automated provisioning reduce manual configuration errors and enable faster deployments.
- **Interference-Aware RF Planning:** Sector placement, channel planning, and power management are designed to minimize RF interference and maximize spectrum efficiency.

PART 1: Rural Site Survey & Feasibility Study

1. Area Description & Market Viability

- **Location & Density:** Sunway Village, Hartbeespoort (real location). Based on municipal boundary mapping, the settlement covers approximately 3.5 km² and has an estimated 1,500 households.
- **Penetration & Downside Modelling:** Target Phase 1 penetration is **20% (300 active subs)** within 6 months. However, the financial model is stress-tested to remain OPEX-positive even at a pessimistic **12% uptake (180 subs)** to ensure survival if competing fibre trenches the main road earlier than anticipated.
- **Income Profile:** Based on comparable semi-urban deployments, sustainable ARPU is modeled at **R350 – R450**. To mitigate price-war risks from future fibre competitors, service differentiation will focus on low-latency gaming optimization and proactive fault detection rather than price undercutting.
- **Elevation & RF LOS Proof:** To counter the dense corrugated iron roofing, we will erect a **30m lattice mast** on leased central high ground. At our maximum cell edge of 2.5 km, the 5.8 GHz first Fresnel zone radius is ~5.7 meters. Assuming an average residential roof height of 4 - 5 meters, the 30m mast provides over 15m of clearance at the midpoint which will achieve sufficient Fresnel clearance for an estimated 90% of residential rooftops. This is subject to a final on-site RF survey validation.
- **LOS vs NLOS Strategy:** While the 30m mast covers 90% of the area, the remaining 10% of households will inevitably suffer from Non-Line-of-Sight (NLOS) conditions due to dense tree cover or topographic dips. For these installs, we will deploy a 3 - 6 meter localized galvanized pole at the customer premises to clear the obstruction. Installations that cannot meet a minimum of -65 dBm RSSI and 15 dB SNR will not be commissioned to protect sector airtime and prevent poor customer experiences. Thus, we will not deploy 2.4 GHz as a fallback due to severe spectrum pollution.
- **Existing Connectivity:** Current options are limited to expensive mobile LTE with poor indoor penetration. There is no existing FTTH (Fibre to the Home) infrastructure.

2. Spectrum Strategy & Interference Mitigation

- **Frequency:** Unlicensed 5 GHz Point-to-Multipoint (PtMP).
- **Interference Mitigation:** To survive the high local noise floor, we will restrict channel widths to **20 MHz** (increasing Power Spectral Density) and utilize DFS (Dynamic Frequency Selection) channels to operate in cleaner airspace. We will enforce a strict ABAB channel reuse pattern across the four sector antennas, shielded by RF armor to mitigate self-interference.
- **PtMP vs Mesh:** We are deploying a PtMP architecture rather than a Wireless Mesh. Mesh networks halve their throughput at every hop and introduce unacceptable latency for voice/gaming. PtMP guarantees deterministic performance from the tower directly to the CPE.

3. Field Survey Methodology

Before final mast placement, a structured RF and infrastructure survey will be conducted:

Tools

- RF spectrum analyser (Ubiquiti AirView)
- GPS mapping tool (QGIS)
- Laser rangefinder
- Drone-based aerial inspection
- LTE signal analyzer for backup connectivity

Data Collected

- Noise floor per 5 GHz channel
- LOS obstruction mapping
- GPS coordinates of candidate tower locations
- Roof height sampling
- Power availability assessment

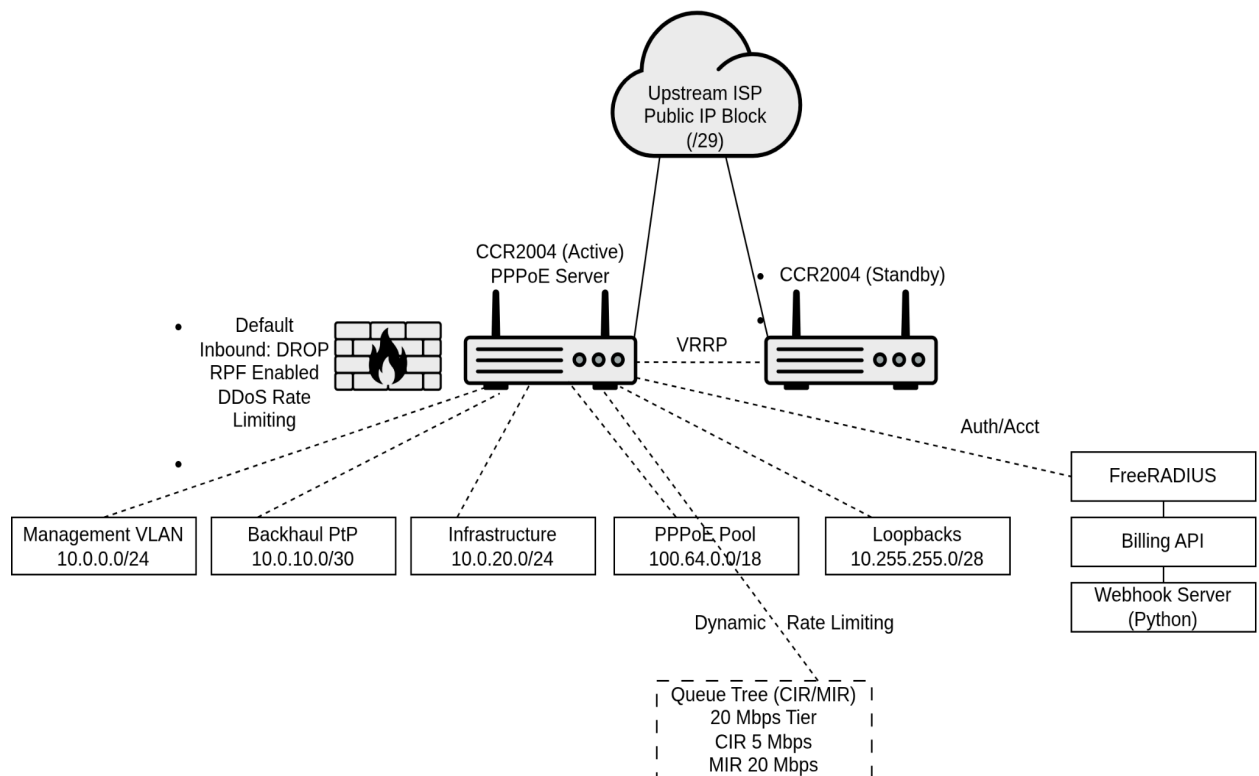
Outputs

- RF heatmap
- Fresnel clearance validation
- Sector orientation optimisation

The Backhaul Link (Weather-Resilient): Given Hartbeespoort's severe summer thunderstorms, a 60 GHz primary link over a multi-kilometer distance is too susceptible to heavy rain fade. Therefore, the primary backhaul will be an **11 GHz Licensed Microwave link** engineered for gigabit capacity. **The 11 GHz link will be engineered for 99.95% availability using ITU-R P.530 rain attenuation models for the North West region.** This is backed up by an automatic **5 GHz unlicensed failover link** utilizing OSPF cost-based routing. Target end-user service availability is designed for 99.5% in Year 1, improving to 99.9% post-fibre augmentation. End-user SLA accounts for access-layer variability beyond backhaul resilience. Furthermore, this deployment assumes existing ECNS/ECS licensing compliance under the ICASA regulatory framework.

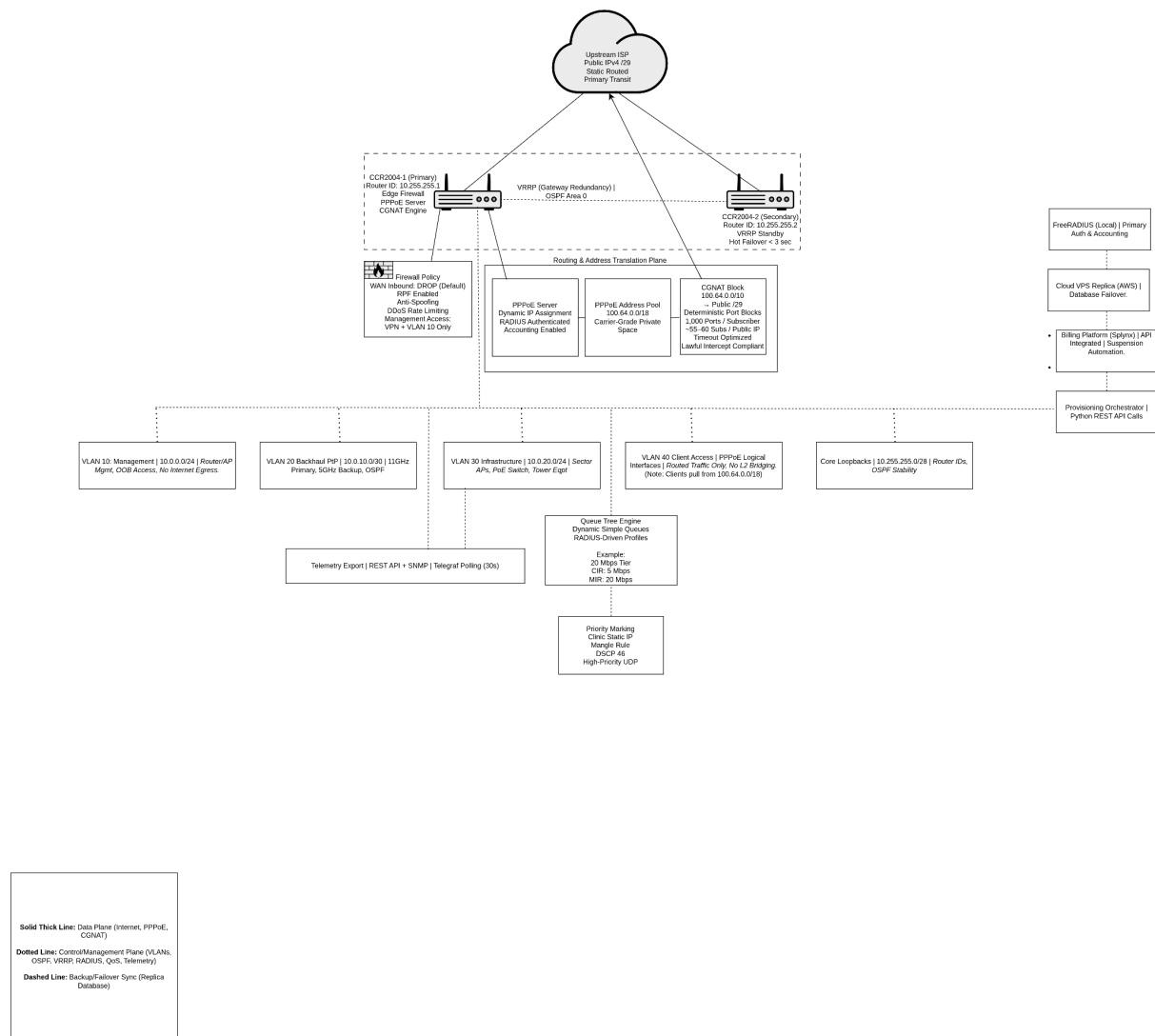
2. Core Network & Zero-Trust Architecture

Core Network Logical Diagram



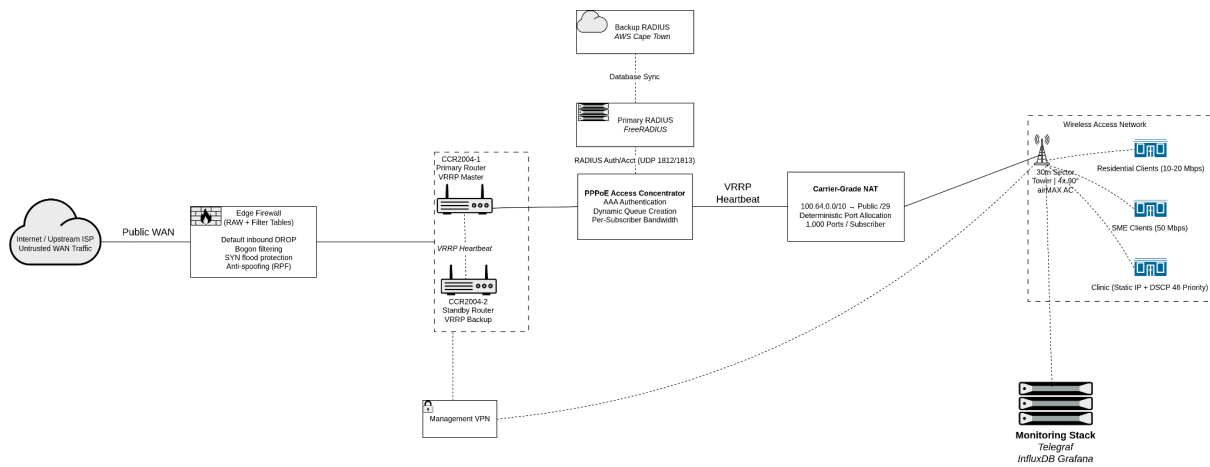
- **Router Redundancy:** Dual MikroTik CCR2004 routers running RouterOS v7 configured with VRRP. If the primary hardware fails, the standby assumes the gateway IP in under 3 seconds.
- **RADIUS Redundancy:** Primary FreeRADIUS server hosted on local hardware, with a lightweight secondary replica in a cloud VPS (e.g., AWS Cape Town) for database failover.
- **CGNAT Scaling & Compliance:** We map the 100.64.0.0/10 block to a public /29 IPv4 pool. By aggressively managing TCP/UDP timeouts and allocating a deterministic block of 1,000 ports per subscriber, we safely house **~55–60 subscribers per public IP** depending on timeout tuning and reserved ports. This deterministic port allocation ensures strict lawful intercept compliance.
- **IP Addressing Plan:** To ensure scalable routing and isolation, we utilize a strictly segmented IPv4 plan:
 - Management VLAN (OOB & Devices): 10.0.0.0/24
 - Backhaul PtP Links: 10.0.10.0/30
 - Infrastructure (APs/Sectors): 10.0.20.0/24
 - PPPoE Client Pool (CGNAT): 100.64.0.0/18
 - Core Loopbacks (OSPF Router IDs): 10.255.255.0/28
 - **OSPF Routing Topology:** All routed infrastructure participates in OSPF Area 0. Access VLAN interfaces are set to passive to prevent rogue adjacency, and no external redistribution is permitted into the IGP.
 - **IPv6 Strategy:** IPv6 dual-stack with delegated /56 per subscriber using DHCPv6-PD will be introduced in Phase 2 to reduce CGNAT load and improve compatibility with modern OTT services.

○ IP Addressing & VLAN Segmentation Diagram



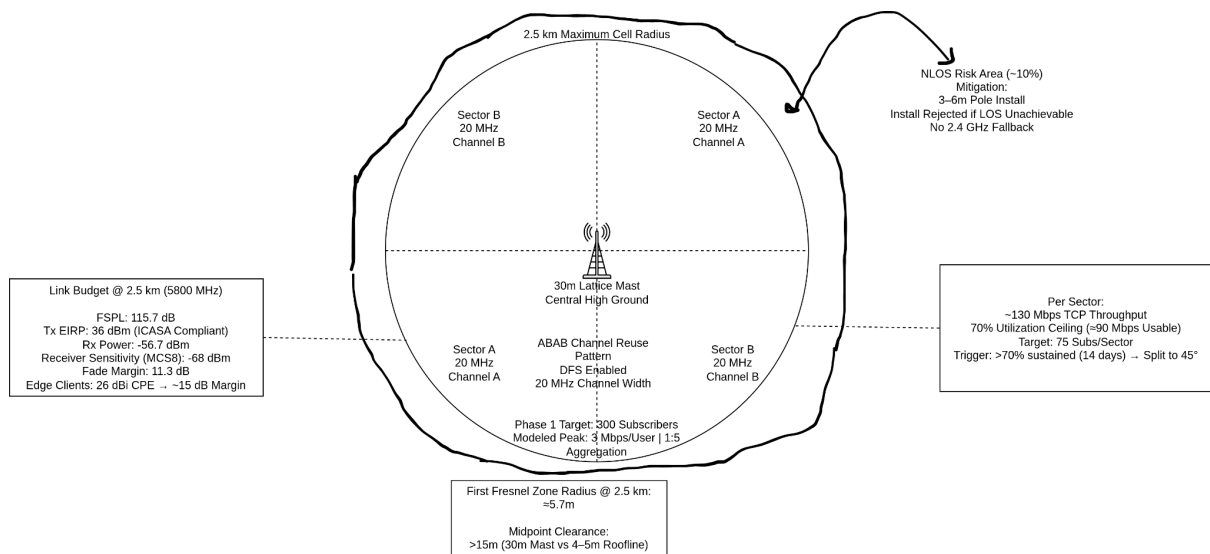
- **Edge Security & Firewall Strategy:** The core CCR router acts as the primary edge firewall. The default inbound policy on the WAN interface is strictly **drop**. Anti-Spoofing (RPF) is enabled. Management plane access (SSH/WinBox) is locked exclusively to the internal Management VLAN and remote VPN subnets. To protect CPU resources, the edge firewall includes RAW table pre-filtering of bogon ranges and SYN flood protection with strict connection-limit thresholds.

○ Security & Traffic Flow Diagram



3. Last-Mile Wireless & Capacity Planning

Tower and Sector Layout



Sector Capacity Math: Each 90° sector operating on 20 MHz (airMAX AC) delivers ~130 Mbps real-world TCP throughput. With a 70% utilization ceiling (≈90 Mbps usable), **each sector is engineered to safely support ~75 residential users at modeled peak load.**

Link Budget Formula @ 2.5 km (5800 MHz):

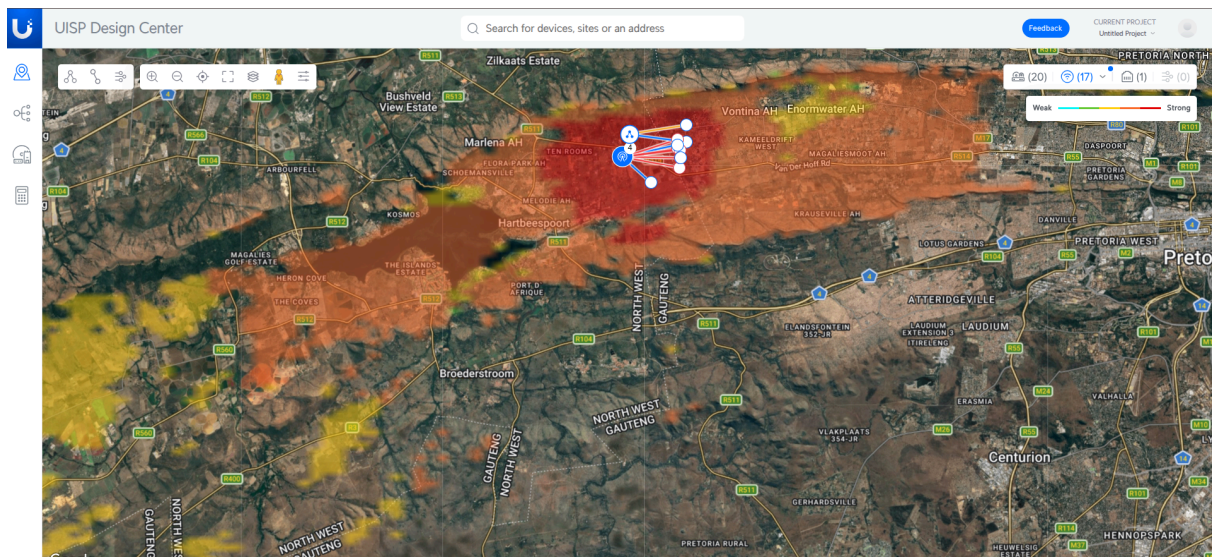
- **FSPL:** 115.7 dB
- **Tx Setup:** Tx Power (15 dBm) + Sector Antenna (21 dBi) = 36 dBm EIRP (Strictly ICASA Compliant).

- **Received Power (Rx):** Tx EIRP (36) - FSPL (115.7) + CPE Rx Gain (23) = **-56.7 dBm**.
- **Fade Margin:** With receiver sensitivity for maximum modulation (MCS8) at -68 dBm, our Rx of -56.7 dBm provides an **11.3 dB Fade Margin**. We will mandate 26 dBi CPE dishes for the furthest 10% of edge clients to push the fade margin closer to the industry-ideal 15 dB.

AP Density & Geographic Coverage:

- **Tower Placement:** A single central 30m mast provides optimal line-of-sight.

Village Coverage Map



- **AP Density Calculation:** To service the 300 Phase-1 households within a 2.5 km radius, 4x 90° APs provide full 360° coverage. At our modeled 20% penetration, this equates to a density of ~75 subscribers per AP, which sits perfectly within the 90 Mbps usable capacity ceiling of a 20 MHz channel.

PART 3: Vendor Selection & Cost Model

1. Vendor Justification

Selected Stack: MikroTik (Core) + Ubiquiti airMAX AC (Edge)

Vendor Stack	Cost per Sub	Interference Handling	API/Automation Support	Verdict
Cambium ePMP	High (R3,500+)	Excellent (GPS Sync)	Good	Too expensive for a R400 ARPU market.
Pure MikroTik	Lowest (R900+)	Poor (TDMA protocol lacks polish)	Excellent	Edge is too susceptible to noise for dense PtMP.
MikroTik + Ubiquiti	Moderate (R1,500)	Very Good (airMAX AC)	Excellent (RouterOS API)	Selected. Maximum API control with reliable RF.

Additional Justification: The Ubiquiti airMAX edge was also selected for its low power consumption (crucial for our solar battery sizing) and its free, centralized UISP Cloud Controller, which is mandatory for our zero-touch provisioning workflow.

2. High-Level CAPEX & ROI Model

- **Total Base Station CAPEX:** R 170,000 (30m Mast, Solar/Batteries, Core Routing, 11 GHz Backhaul, Sectors).
- **CPE CAPEX (300 units @ R1,500 installed):** R 450,000.
- **Total Project CAPEX: R 620,000.**
- **OPEX Assumptions:** Assumes wholesale IP transit cost of R90/Mbps (R27,000/mo for 300 Mbps). *Note: This wholesale pricing assumption is conservative; exact pricing will be confirmed during the vendor engagement phase.* Total OPEX (bandwidth, site lease, power) is estimated at R35,000/mo.

- **ROI Timeline:** At 300 subs generating R120,000 MRR, minus R35,000 OPEX, the network generates R85,000 gross profit monthly. **Breakeven on total CAPEX is expected within 8–12 months**, depending on churn and installation velocity.

Financial Sensitivity & Downside Protection Model

Uptake Scenario	Active Subs	Est. MRR	Est. OPEX	Net Gross Profit	Break-Even Impact
Pessimistic (12%)	180	R 72,000	R 35,000	R 37,000	~16 Months
Target Phase 1 (20%)	300	R 120,000	R 35,000	R 85,000	~8 Months

PART 4: Authentication & Risk Mitigation

1. Authentication & QoS

We deploy PPPoE over VLAN-based IP transport to natively prevent wireless broadcast storms and provide strict accounting updates for billing.

- **Dynamic QoS & Rate Limiting Model:** Upon PPPoE login, RADIUS dynamically spawns Simple Queues capping the user at their paid tier (e.g., 10 Mbps/20 Mbps/50 Mbps). We utilize a CIR/MIR (Committed/Maximum Information Rate) model. A 20 Mbps tier has a MIR of 20 Mbps and a CIR of 5 Mbps, guaranteeing baseline performance during peak congestion.
- **Fair Usage Policy (FUP):** Uncapped tiers are subject to a sliding-scale FUP. If a residential user exceeds 500GB in a 30-day rolling window, RADIUS CoA (Change of Authorization) automatically drops their MIR by 50% during peak hours (6 PM - 10 PM) to protect sector airtime for standard users.
- **Voucher-Based Access:** Not applicable. As a fixed-wireless residential ISP focused on stable ARPU, we do not deploy public Wi-Fi hotspots, eliminating the overhead of voucher generation and transient user support.
- **Priority Marking:** Traffic originating from the static IP of the local clinic/school is marked via MikroTik Mangle at ingress, tagged with DSCP 46, and prioritized in the global queue tree to protect telemedicine UDP packets during peak hours.
- **Subscription Billing Integration:** The FreeRADIUS server integrates directly via API with an ISP billing platform (e.g., Splynx). When a customer's prepaid account runs out of funds, the billing system instructs RADIUS to automatically assign a 'Suspended' address list profile, redirecting the user to a payment gateway without manual intervention.

2. Expanded Risk Register

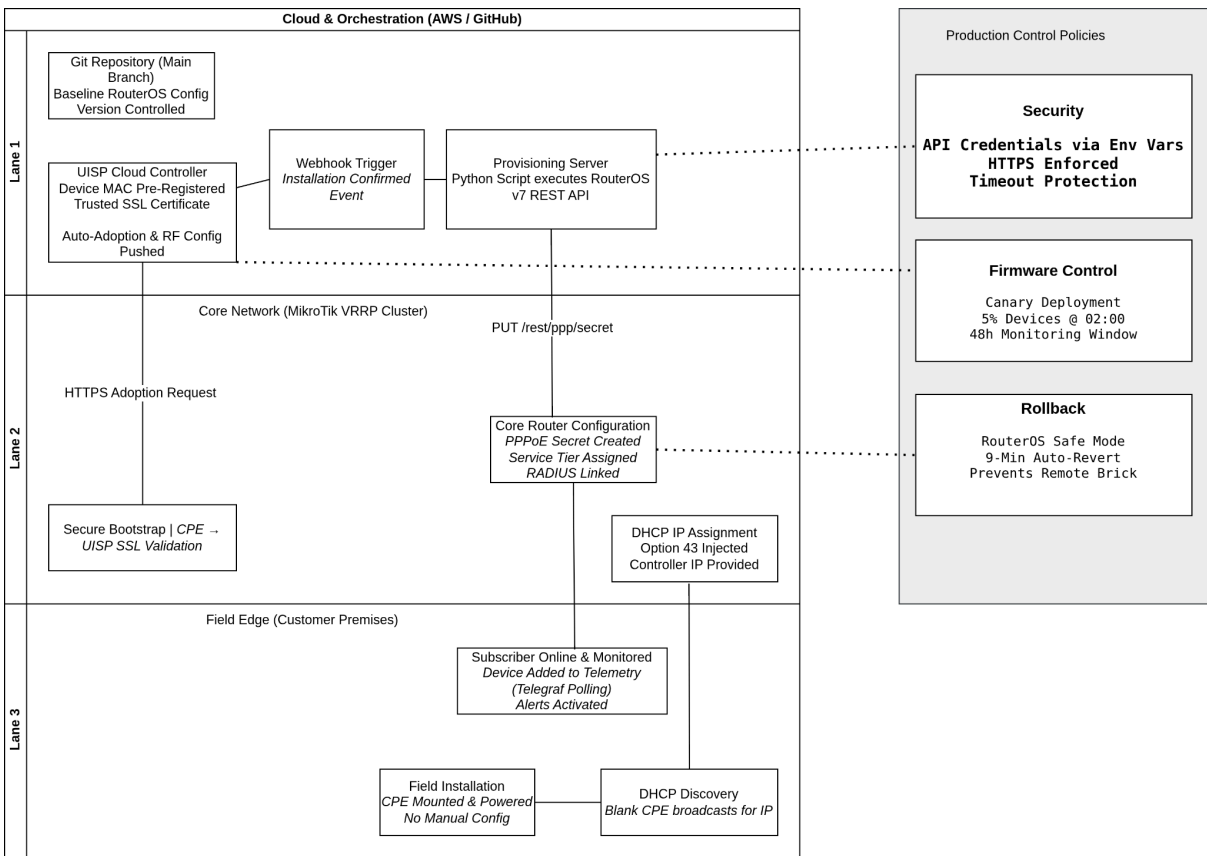
Risk Event	Impact	Mitigation Strategy
Stage 6 Load Shedding	Tower goes offline.	At an estimated 300W total tower draw, a 400Ah battery bank at 48V provides ~19.2 kWh of usable energy, yielding ~32–36 hours of autonomy after inverter losses.
Subscriber Growth Exceeds Sector Capacity	Congestion & churn	Predefined trigger: if sector utilization >70% sustained for 14 days, deploy additional sector.
11 GHz Backhaul Failure	Total isolation.	OSPF auto-failover to 5GHz backup link within 3 seconds. Tertiary LTE router installed for out-of-band management access.
Fibre Competitor Enters	High churn rate.	Prepaid, month-to-month contracts. CAPEX breaks even at 12% penetration.
CGNAT IP Blacklisting	Subs blocked from PSN.	Deterministic port allocation allows exact isolation of the abusing subscriber.

PART 5: Zero-Touch Rural Deployment Plan

To rapidly deploy Sunway Village without requiring Senior Engineers on-site, we utilize a strict GitOps and Zero-Touch lifecycle.

1. The Lifecycle Workflow

Zero-Touch Deployment Workflow



1. **Cloud Pre-Registration:** Device MAC addresses and serial numbers are pre-registered in the Ubiquiti UIISP Cloud Controller.
2. **Git-Based Config:** The baseline MikroTik configuration template is committed to the **main** branch of our repository.
3. **Field Installation:** A junior technician physically mounts the CPE at the customer's roof and powers it on.
4. **DHCP Option 43 Auto-Adoption:** The CPE receives an IP via DHCP. The MikroTik injects DHCP Option 43, telling the completely blank CPE the exact IP address of the UIISP Cloud Controller.
5. **Secure Bootstrap:** The CPE reaches out to UIISP. The controller presents a trusted SSL certificate, preventing man-in-the-middle attacks, and pushes the RF configuration.

6. **RADIUS Automation (Webhook):** UISP triggers a webhook to our orchestration server, executing a Python script that automatically provisions the PPPoE account on the core router.

2. Remote Firmware & Rollback Strategy

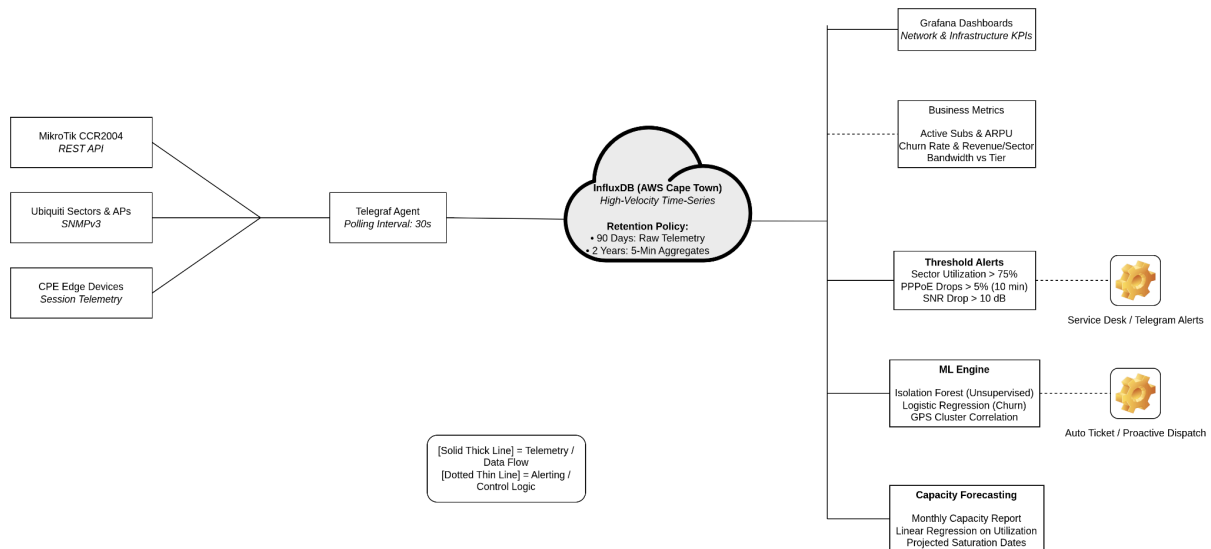
- **Staged Firmware Rollout:** Firmware updates are automated via a "Canary" deployment—pushed remotely to 5% of non-critical edge CPEs at 2 AM. If monitoring detects no anomalies over 48 hours, the remaining 95% are patched.
- **Configuration Rollback:** If a pushed configuration severs management plane connectivity, RouterOS **Safe Mode** automatically reverts the running config to the previous state after a 9-minute timeout, preventing a remote "brick" scenario.

3. Python Automation Deliverable: PPPoE Provisioning This script satisfies the requirement for API automation, demonstrating how our orchestration server automatically provisions a new customer on the MikroTik Core Router upon successful physical installation.

PART 6: Data-First Monitoring & Analytics

The network will be operated using a telemetry-driven monitoring model that combines network engineering with data science.

Monitoring Diagram



1. The Core Architecture & Data Retention

A central Telegraf agent polls the MikroTik REST API and Ubiquiti SNMP agents every 30 seconds. This high-velocity telemetry is pushed directly into an InfluxDB time-series database and visualized on Grafana. To preserve visibility during tower-level power or backhaul failures, the entire monitoring stack is hosted off-site in a cloud VPS (e.g., AWS Cape Town).

- **Data Retention & Storage Strategy:** To ensure POPIA compliance and optimize storage costs, 30-second high-velocity telemetry is retained for 90 days for short-term anomaly detection. Downsampled 5-minute aggregates are retained for 2 years for seasonal capacity planning. Billing and usage data are retained strictly per POPIA requirements.

2. Data Model & Feature Engineering Raw telemetry is transformed into actionable features before entering the ML pipeline:

- **Features per AP:** Avg RSSI, SNR variance, channel utilization %, PPPoE session drops/hour, throughput delta vs tier, retransmission rate.
- **Features per Sector:** 95th percentile utilization, client count delta, median latency deviation.
- **Features per Subscriber:** Usage variance, connection stability index.
- **Normalization:** Features are normalized using rolling z-score normalization over a 30-day window before feeding into the anomaly detection model.

3. Multi-Layered Alerting & Automated Anomaly Detection

- **Layer 1 (Deterministic Alerting):** Hard thresholds trigger immediate service desk alerts: Sector utilization > 75%, PPPoE drops > 5% in 10 minutes, or an SNR drop > 10 dB sustained.
- **Layer 2 (Isolation Forests):** Leveraging Isolation Forests against the InfluxDB data, anomalies automatically create internal service desk tickets correlating affected subscribers by GPS cluster.
- **Model Governance:** To maintain predictive accuracy in a dynamic RF environment, models are retrained weekly using a 70/30 time-based split for validation. Alert suppression thresholds are enforced to prevent noise storms during known widespread outages (e.g., regional load shedding), and the target False Positive Rate (FPR) is strictly maintained at < 5%.
- *Model performance metrics (Precision, Recall, F1-score) are logged monthly to detect drift and degradation.*

4. Predictive Churn Modeling & Capacity Forecasting

- **Predictive Churn:** Combining network telemetry with business outcomes, we deploy a logistic regression model to estimate churn probability. Using subscriber usage volatility, latency complaints, and service desk ticket frequency as primary features, subscribers scoring a churn probability > 0.65 trigger automated, proactive retention outreach before they cancel their service.
- **Capacity Forecasting:** Capacity planning reports are auto-generated monthly from InfluxDB, projecting sector saturation dates using linear regression on utilization trends.

PART 7: Rural Power & Physical Resilience

To mitigate the chance of hardware getting destroyed by a highveld thunderstorm or stolen, the base station will be hardened to industrial standards:

- **Solar Backup & Battery Sizing:** At an estimated 300W total tower draw, a 400Ah battery bank at 48V provides ~19.2 kWh of usable energy, yielding ~32-36 hours of autonomy after inverter losses. This is sustained by a 1.2kW solar array configured with redundant MPPT charge controllers.
- **Lightning & Surge Protection:** Sunway Village experiences severe lightning. We will install a dedicated copper earthing spike array (< 5 Ohms). Every Cat6a cable is terminated through an armored Ethernet Surge Protector (e.g., Ubiquiti ETH-SP-G2) at the top of the tower and again before entering the cabinet.
- **Environmental Control:** The outdoor router enclosure is IP65-rated to prevent dust and water ingress. An IoT temperature sensor feeds data to Grafana; if internal temps exceed 45°C, dual exhaust fans are automatically triggered.
- **Theft Mitigation:** The batteries and smart inverters are housed in a heavy-gauge, tamper-proof steel cabinet bolted to a concrete plinth. A dedicated IP CCTV camera is mounted at 3 meters on the mast, recording locally and sending motion-triggered alerts to the engineering Telegram group.
- **Occupational Health & Safety (OHS):** All physical tower access and maintenance will require a strict dual-person policy to ensure safety compliance and mitigate operational risk.

PART 8: Scalability Roadmap

To easily handle 2x subscriber growth (300 → 600):

- **Trigger-Based Expansion:** When any existing 90° sector hits 70% sustained utilization, we split the cell by deploying two additional 45° horn antennas.
- **Long-Term Backhaul:** Once active subscribers cross 500, the 11 GHz microwave link will be augmented with a direct fibre handoff at the tower base, repurposing the microwave as a high-capacity redundant path.
- **Geographic Expansion:** To add neighboring villages, we will utilize the Sunway tower as a high-site repeater, deploying a dedicated PtP (Point-to-Point) link to feed a new micro-pop tower in the adjacent settlement.
- **Licensed Spectrum Migration:** As the 5 GHz unlicensed band reaches saturation and the subscriber base scales, last-mile sectors will be systematically migrated to licensed 3.5 GHz (WiMAX/LTE) or 6 GHz to guarantee interference-free airtime.

PART 9: Final Architecture Summary

The proposed Sunway Village WISP architecture is designed to deliver scalable broadband connectivity to underserved rural communities using a resilient, cost-efficient wireless infrastructure.

The architecture combines:

- Licensed microwave backhaul for high availability
- Sector-based 5 GHz PtMP wireless distribution
- PPPoE-based subscriber management with RADIUS authentication
- Carrier-grade NAT for IPv4 conservation
- Segmented VLAN architecture for operational security
- Automated monitoring and telemetry-driven operations

The system is engineered to support **300 initial subscribers with scalable growth to 600+ users** without requiring fundamental architectural changes.

PART 10: Design Validation & Operational Readiness

1 Pre-Deployment Validation

Before physical deployment, the network design will undergo validation to ensure predicted performance aligns with operational requirements.

Validation methods include:

- RF coverage simulation using **Ubiquiti UISP Design Center**
- Link budget calculations for tower-to-node backhaul connections
- Capacity modelling based on projected subscriber density
- Channel interference analysis for 5 GHz spectrum and DFS channel availability
- Configuration testing within a controlled lab environment before field deployment

These validation steps reduce deployment risk and ensure the network performs as expected under real operating conditions.

2 Deployment Acceptance Testing

After installation, each network component will undergo **acceptance testing** to verify correct operation.

Validation includes:

- **Signal Strength Verification**
Confirm that client RSSI values remain above -65 dBm in coverage zones.
- **Throughput Testing**
Conduct performance tests using tools such as iperf to confirm expected bandwidth capacity.
- **Roaming Validation**
Validate subscriber connectivity stability across sector coverage zones.
- **Latency Measurement**
Measure round-trip latency to the gateway and upstream provider.
- **Packet Loss Monitoring**
Verify packet loss remains below acceptable thresholds during normal operation.

3 Operational Monitoring Metrics

To maintain a **data-driven operational model**, the following key metrics will be continuously monitored.

Metric	Target
Client RSSI	-65 dBm or better
Client latency	<20 ms to gateway
Packet loss	<1%
Sector utilization	<70%
Backhaul link uptime	99.9%

Telemetry is collected via Telegraf from MikroTik and Ubiquiti devices and stored in InfluxDB, while Grafana provides operational dashboards and alerting. UISP is used primarily for device management and provisioning.

PART 11. Engineering Assumptions

- On-site RF survey required before final mast placement
- Wholesale IP pricing subject to supplier negotiation
- Final microwave link engineered post-path profile analysis

Conclusion

The proposed architecture prioritizes reliability, scalability and operational visibility. The platform uses a combination of sector-based fixed wireless access, segmented network design, automated provisioning and a telemetry-driven monitoring stack. This allows it to scale from the initial deployment phase to full village coverage while maintaining predictable performance and manageable operational overhead. This approach ensures the infrastructure can grow alongside subscriber demand without requiring fundamental architectural changes.

Future Engineering Enhancements

Future iterations of the network would include:

- Migration to IPv6 dual-stack with DHCPv6-PD
- Deployment of additional micro-POPs for neighboring settlements
- Integration with NetBox for automated IPAM and infrastructure inventory
- Automated configuration validation using CI pipelines before production rollout